

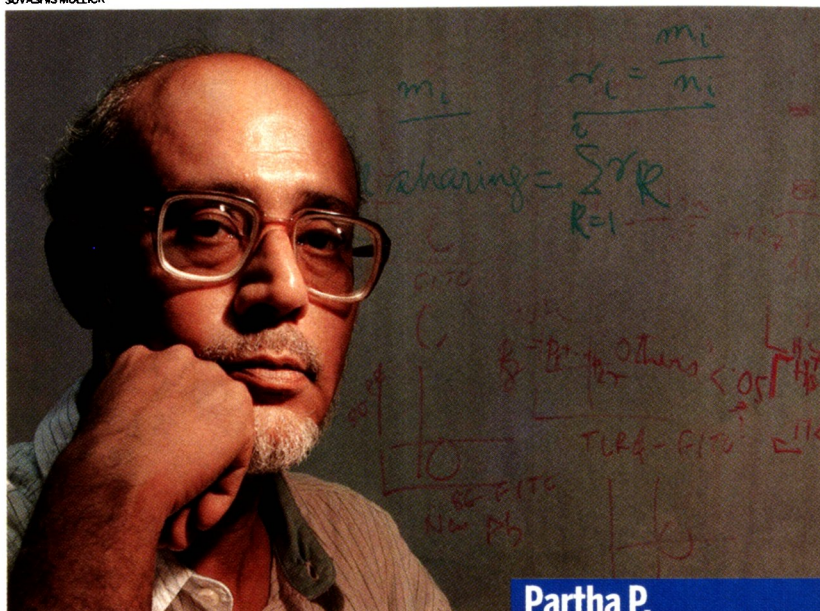
Institute (ISI), Kolkata, Central Drug Research Institute (CDRI), Lucknow, Institute of Microbial Technology (IMTECH), Chandigarh, Industrial Toxicological Research Centre (ITRC), Lucknow and Indian Institute of Chemical Biology (IICB), Kolkata.

The team first collected blood samples of 2,000 people from 55 populations representing the four major linguistic groups: Austro-Asiatic, Tibeto-Burman, Indo-European and Dravidian before splicing their genes and looking for tell-tale variations. In doing so the project established the first large-scale database of genomic diversity in the Indian population. Samir Brahmachari, a geneticist of repute and CSIR's director-general says, "India is among the first country to do a diseases-specific, drug-response mapping on large populations which gives us risk analysis and benefits." Apart from reinforcing what Brahmachari calls the "genetic mosaic" of India, the study also showed that the risk of susceptibility to a disease was more specific to a geographical region rather than to caste, creed or linguistic affinities. Reviewing the consortium's work, Evelyne Heyer, geneticist at the French Natural History Museum, said, "This study forms a valuable first step in building our knowledge of the genetic diversity in India."

While the consortium released its preliminary findings last year, since then individual laboratories have come out with a host of important findings. At the CDRI, geneticist Saman Habib focused on the more lethal form of cerebral malaria caused by the *Plasmodium falciparum*. Studies were conducted on people who lived in the regions where malaria was endemic. Analysing the samples and also from control groups, Habib found a "clear indication of variety of genes affecting the susceptibility of the individual towards this deadly form of malaria including the severity of the attack". One pointer: Health officials would be well-advised to double their efforts in regions with such susceptibility.

At the IGB, Shantanu Sengupta is coming out with surprising findings about cardiovascular diseases and genetic factors. Contrary to popular notions, vegetarians in India have a high susceptibility to heart diseases. That is now found to be due to a deficiency of Vitamin B12. But there was a significant variation of this phenomenon between North

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Partha P. MAJUMDER

Head of Human Genetics Unit, ISI

and South India. The study found that Northern Indians who were also vegetarian had a much higher degree of susceptibility to heart problems than their southern brethren because of a genetic variation that made them vulnerable. The deficiency in Vitamin B12 and the genetic variation increased the level of an amino acids associated as a risk factor for cardiovascular disease.

In a laboratory not far from Sengupta, Balaram Ghosh, Head of the Genomics and Molecular Medicine Unit, IGB is proving just how useful the emerging field known as pharmacogenomics could be. Researching on the cause of asthma in India, his team, after examining over 700 patients, found that many genetic variations that are known to make people susceptible to asthma among

“South Asia played a pivotal role in the out-of-Africa colonisation and dispersal of modern humans.”



SANJAY SONKAR

“There is a clear indication of a variety of genes affecting the susceptibility of the individual towards lethal cerebral malaria.”

Saman HABIB

Scientist, CDRI